Thanakrit Jomhong

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Education

Mahidol University Aug 2024 – May 2028
Bachelor of Engineering Program in Computer Engineering *GPA: 3.90*

Experience

Western Digital, Internship Jun 2026 – Aug 2026
Manufacturing Engineer(Machine Learning / Computer Vision), AVI Vision Enhancement

- Built a YOLO instance segmentation pipeline for an Automated Visual Inspection (AVI) system on a live Genesis production line, targeting false-reject reduction in the ML re-inspection workflow.
- Constructed a labeled defect dataset by hand-annotating 1,500+ production images across 6 defect classes using open-source annotation tooling. established labeling documentation standards to ensure annotation consistency and enable dataset expansion.
- Integrated the model into the existing inspection pipeline, reducing manual review workload by 64%.

Head of Sponsorship Coordinator, ClickCamp16 May 2025 – Dec 2025
Mahidol Computer Engineering

- Led sponsorship strategy and relationship management for a university computing camp, secured 130,000 THB in funding.
- Contacted 200+ companies and built structured outreach-tracking workflows to manage the full sponsorship pipeline.

Projects

Banana Ripeness Classification | ResNet34, PyTorch, scikit-learn

- Built a proprietary dataset from scratch by photographing bananas daily over 3+ months across their full ripeness lifecycle, capturing natural progression from fresh to rotten under real conditions.
- Iterated on class taxonomy through multiple relabeling experiments, testing different ripeness stage boundaries to maximize inter-class separability, improving F1-score to 87.5%.

HR ChatBot | n8n selfhost, Docker, Supabase (PostgreSQL), LLM, RAG

- Developed a cross-platform chatbot to automate HR workflows such as leave requests and benefit inquiries through the LINE Messaging API.
- Built a RAG pipeline over HR policy documents with LLM-based intent extraction, handling multiple intent categories end-to-end.

EnergyPulse | Python, SQL, SQLite, dbt, Apache Airflow, BigQuery

- Developed a data ingestion prototype processing 100,000 synthetic energy records in 0.194 seconds in a local in-memory SQLite benchmark.
- Implemented primary-key-based idempotent replay, reprocessing 10,000 existing records with zero additional rows in the prototype experiment.
- Evaluated SQL query optimization through indexed versus non-indexed execution, measuring a 99.9% reduction in median query latency for a selective analytical query.
- Tested data-quality validation against 1,000 deliberately injected invalid records in a 100,000-record synthetic dataset.
- Designed an extensible ETL/ELT architecture for incremental ingestion, automated data testing, workflow orchestration, and analytics-ready warehouse modeling.

Technical Skills

ML / CV: PyTorch, Ultralytics YOLO, OpenCV, scikit-learn, Pandas, NumPy
Systems & Tools: Docker, Git/GitHub, FastAPI, Linux, Supabase, Vercel, n8n
Data Engineering & Analytics: ETL/ELT Pipelines, Data Cleaning & Validation, SQL Query Optimization, Relational Database Design, Data Modeling, API Data Ingestion

Awards

2nd Place Winner, Super AI Engineer SS5 (AI Innovator Track, Western Region) Summer 2025

Gold Medal, Onsite Presentation, Data Science Project Contest 2023, CMU Sep 2023